2021-22

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SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMKUR.

(A Constituent College of Sri Siddhartha Academy of Higher Education, Agalakote, Tumkur.)

B.E. SEMESTER END EXAMINATION – JAN 2023

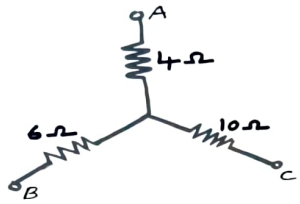
EC3TH3 : NETWORK THEORY

SEM: III

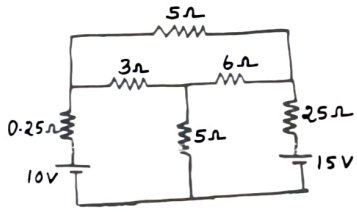
TIME: 3.00 Hrs **MAX MARKS: 100**

NOTE: Answer any five full questions selecting one full question from each choice.

		M	CO	BL
1.a)	Illustrate Kirchoff's laws with example	6	4	2
b)	Replace the star connection by an equivalent delta connection	6	4	



c)	Determine currents in all branches using loop method	8	4	2



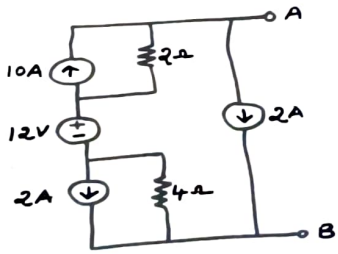
$8.25I_1 - 5I_2 - 3I_3 = -10$

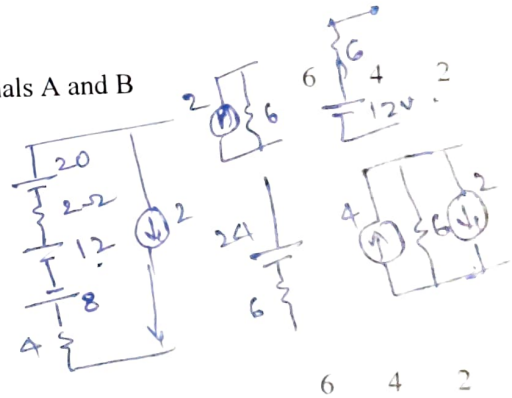
$-5I_1 + 36I_2 - 6I_3 = 15$

$-3I_1 - 6I_2 + 14I_3 = 0$

OR

2.a)	Reduce the network shown to a single voltage source between terminals A and B	6	4	2



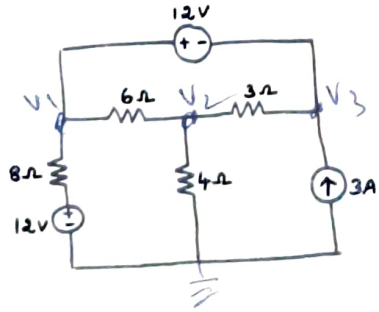


b)	Illustrate Delta to Star transformation	6	4	2
c)	Determine the voltage across the 4Ω resistor and 3A current source in the network using node voltage method	8	4	2

$V_A = 68V, V_2 = 40$

$V_C = 56V$

$V_{4\Omega} = V_2 = 40V$



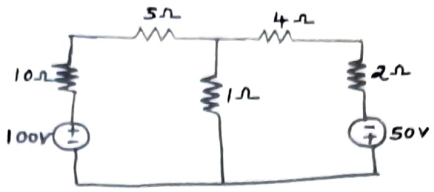
$V_1 - V_3 = 12$

$\frac{V_2 - V_1}{6} + \frac{V_2 - V_3}{3} + \frac{V_2}{4} = 0.$

$\frac{V_1 - 12}{8} + 3 + \frac{V_1 - V_2}{6} + \frac{V_3 - V_2}{3} = 0.$

3.a) Find the current through 1Ω resistor using Superposition theorem. 6 1 1

6.756 A
7.207 A
R_{eq} = 6.937Ω
I_2 = 5.405
$I_{1\Omega}$ = −1.3515 (A to B)

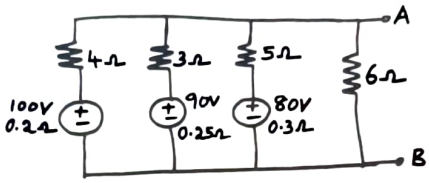


b) Illustrate maximum Power transfer theorem with pure resistive load 6 1 2

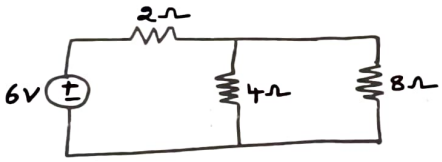
c) State and prove Thevenin's theorem. 8 1 3

OR

4.a) Determine the current in 6Ω resistor using Millman's theorem. 6 1 2



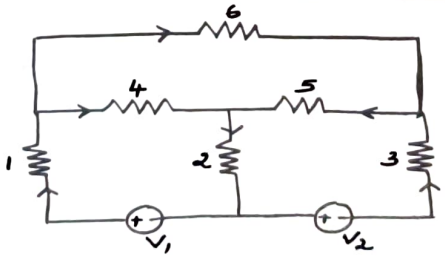
b) Determine the current through 8Ω resistor .Apply Reciprocity theorem and compare the two currents. 6 1 2



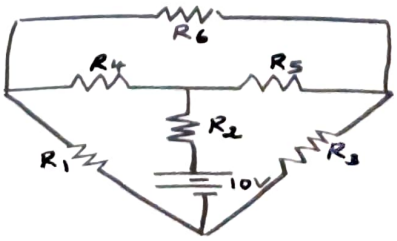
c) State and prove Norton's theorem 8 1 3

5.a) Define Bandwidth, Selectivity and Q-factor from the frequency response of series resonance 6 2 1

b) Write the complete incidence matrix for the network shown. 6 2 1



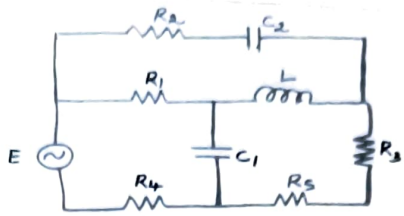
c) For the network shown obtain the tieset schedule by taking 2,4,5 as twigs and write the equilibrium equations. 8 2 1



OR

6.a) An impedance coil having a resistance of 25 Ω and an inductance of 25 mH is connected in parallel with a variable capacitor. For what value of C will the circuit be resonant if a 90 V, 400Hz source is applied. 6 2 1

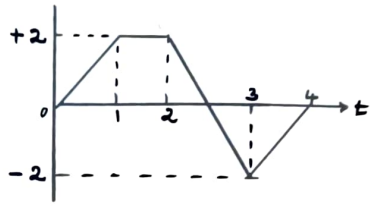
b) Draw the Dual of the network shown. 6 2 2



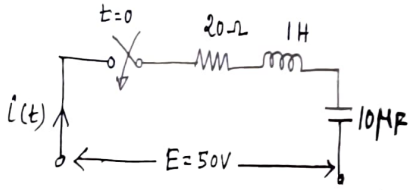
c) Show that the resonant frequency is the geometric mean of the two half power frequencies. 8 2 2

7.a) Verify Initial value theorem for $f(t) = 10\,e^{5t}$ 6 1 2

b) Find the laplace transform for the waveform shown. 6 1 1



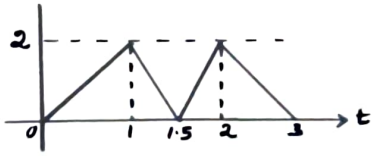
c) Find i(o+) ,di/dt (o+) and d^2i/dt^2(o+) if the switch is closed at t= 0 8 3 1



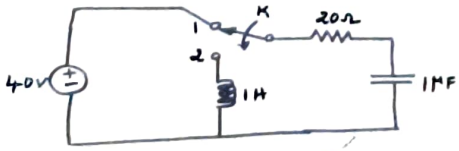
OR

8.a) Verify the Final value theorem for $f(t) = 15(1- e^{-5t})$ 6 1 2

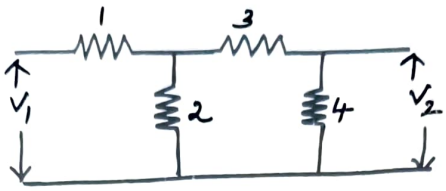
b) Find the laplace transform for the waveform shown. 6 3 1



c) For the circuit shown the switch k is changed from position 1 to 2 at t=0 , steady state having been reached before switching Evaluate i(o+) ,di/dt (o+) and d^2i/dt^2(o+). 8 1 1



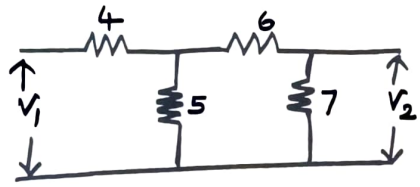
9.a)	Illustrate the Z parameter equivalent of the two port network.	6	2	2
b)	Determine the Y parameters of the network shown.	6	2	2



c)	Find the relationship between Z and Y parameters	8	2	1

OR

10.a)	Illustrate the H-parameter equivalent of the two port network.	6	2	2
b)	Determine the Z parameters of the network shown.	6	2	2



c)	Find the relationship between Z and Y parameters	8	2	1

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